

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED: CO-4 - Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Pseudocode Writing Task (Algorithm Design)

Weightage: 20%

QUESTIONS: Total Marks: 25

Code of Conduct:

I Adhil Bijukumar (192411133) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Adhil Bijukumar

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs

Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
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Question 1: XML Parsing Algorithm

Write a detailed pseudocode algorithm to parse an XML document that contains structured data such as user or product details.

Pseudocode:

```

BEGIN
  DECLARE xmlFile, xmlDocument, rootElement
  DECLARE nodeList, currentNode, childNode
  xmlFile = OPEN("data.xml")
  IF xmlFile NOT FOUND THEN
    DISPLAY "Error: XML file not found"
    EXIT
  END IF
  xmlDocument = PARSE(xmlFile)
  rootElement = GET_ROOT_ELEMENT(xmlDocument)
  nodeList = GET_CHILD_NODES(rootElement)

  FOR EACH currentNode IN nodeList DO
    IF currentNode.TYPE == "ELEMENT_NODE" THEN

      // Step 3: Extract required data (e.g. user/product record)
      DECLARE record = EMPTY MAP
      FOR EACH childNode IN GET_CHILD_NODES(currentNode) DO
        IF childNode.TYPE == "ELEMENT_NODE" THEN
          record[childNode.TAG_NAME] = GET_TEXT_CONTENT(childNode)
        END IF
      END FOR

      // Step 4: Handle nested/hierarchical child elements recursively
      IF HAS_CHILD_ELEMENTS(currentNode) THEN
        CALL PARSE_NODE(currentNode) // recursive traversal for nested structure
      END IF

      // Step 5: Display output in readable format
      DISPLAY "Record: " + currentNode.TAG_NAME
      FOR EACH (key, value) IN record DO
        DISPLAY "  " + key + " : " + value
      END FOR

    END IF
  END IF

```

```

END FOR

CLOSE(xmlFile)
END

FUNCTION PARSE_NODE(node)
  FOR EACH child IN GET_CHILD_NODES(node) DO
    IF child.TYPE == "ELEMENT_NODE" THEN
      DISPLAY " Sub-element: " + child.TAG_NAME + " = " + GET_TEXT_CONTENT(child)
      IF HAS_CHILD_ELEMENTS(child) THEN
        CALL PARSE_NODE(child) // handle deeper hierarchy
      END IF
    END IF
  END FOR
END FUNCTION

```

Question 2: MVC Workflow

Write a pseudocode representation of the MVC (Model-View-Controller) workflow in a web application.

Pseudocode:

```

BEGIN
  // Step 1: User sends a request
  userRequest = RECEIVE_HTTP_REQUEST(FROM client)

  // Step 2: Request reaches the Controller
  controller = ROUTE_REQUEST(userRequest)

  IF controller NOT FOUND THEN
    DISPLAY "404 - Page Not Found"
    EXIT
  END IF

  // Step 3: Controller processes the request
  requestData = EXTRACT_PARAMETERS(userRequest)
  action = DETERMINE_ACTION(controller, requestData)

  // Step 4: Controller interacts with Model
  IF action REQUIRES_DATA THEN
    model = LOAD_MODEL(action.entityName)
    resultData = model.FETCH_DATA(requestData) // Model queries the database
    resultData = model.VALIDATE_AND_PROCESS(resultData)
  END IF

  IF action REQUIRES_UPDATE THEN
    model.SAVE_DATA(requestData) // Model updates the database
  END IF

```

```

// Step 5: Controller selects the View
view = SELECT_VIEW(action.viewName)

// Step 6: View generates the response using Model data
renderedOutput = view.RENDER(resultData)

// Step 7: Response sent back to the user
SEND_HTTP_RESPONSE(TO client, renderedOutput)
END

```

Question 3: JSP Request Flow

Write a detailed pseudocode algorithm to explain the request-response flow in a JSP-based web application.

Pseudocode:

```

BEGIN
// Step 1: Client initiates request
clientRequest = SEND_REQUEST(url = "page.jsp")

// Step 2: Request reaches the Web Server / Servlet Container
server = RECEIVE_REQUEST(clientRequest)

IF requestedResource == "*.jsp" THEN

    // Step 3: Check if JSP is already compiled/translated
    IF JSP_NOT_COMPILED OR JSP_MODIFIED THEN
        servletCode = TRANSLATE_JSP_TO_SERVLET(page.jsp)
        COMPILE(servletCode)
    END IF

    // Step 4: Servlet container loads and executes the servlet
    servletInstance = LOAD_SERVLET(compiledServlet)

    // Step 5: Interaction with backend components
    IF request NEEDS_DATA THEN
        dbConnection = OPEN_DATABASE_CONNECTION()
        queryResult = dbConnection.EXECUTE_QUERY(sqlStatement)
        CLOSE(dbConnection)
    END IF

    IF request USES_OTHER_SERVLET THEN
        businessResult = CALL_SERVLET(helperServlet, requestData)
    END IF

    // Step 6: Generate dynamic content
    dynamicContent = MERGE(HTML_TEMPLATE, queryResult)

```

```

// Step 7: Send generated response back to client
response = BUILD_HTTP_RESPONSE(dynamicContent)
SEND_RESPONSE(TO client, response)

ELSE
    DISPLAY "Error: Unsupported resource type"
END IF
END

```

Question 4: Database Retrieval

Write a pseudocode algorithm to retrieve data from a database.

Pseudocode:

```

BEGIN
    DECLARE connection, query, resultSet, row

    // Step 1: Establish database connection
    connection = CONNECT_TO_DATABASE(host, username, password, dbName)

    IF connection FAILED THEN
        DISPLAY "Error: Unable to connect to database"
        EXIT
    END IF

    query = "SELECT * FROM table_name WHERE condition"
    resultSet = connection.EXECUTE_QUERY(query)

    IF resultSet IS EMPTY THEN
        DISPLAY "No records found"
    ELSE
        // Step 3: Process the result set
        WHILE resultSet HAS_NEXT_ROW DO
            row = resultSet.GET_NEXT_ROW()

            // Step 4: Display retrieved data
            DISPLAY row.column1, row.column2, row.column3
        END WHILE
    END IF

    // Step 5: Close connection and release resources
    resultSet.CLOSE()
    connection.CLOSE()
END

```

Question 5: PHP Validation

Write a pseudocode algorithm for validating user input in a web form using PHP concepts.

Pseudocode:

```
BEGIN
DECLARE name, email, password, errors = EMPTY LIST
IF FORM_SUBMITTED (method = "POST") THEN
    name = GET_POST_VALUE("name")
    email = GET_POST_VALUE("email")
    password = GET_POST_VALUE("password")
    IF name IS EMPTY THEN
        errors.ADD("Name field is required")
    END IF

    IF email IS EMPTY THEN
        errors.ADD("Email field is required")
    END IF

    IF password IS EMPTY THEN
        errors.ADD("Password field is required")
    END IF

    // Step 3: Validate email format
    IF email IS NOT EMPTY AND NOT MATCHES_PATTERN(email, EMAIL_REGEX) THEN
        errors.ADD("Invalid email format")
    END IF

    // Step 4: Validate password format (e.g. min 8 chars, letter + number)
    IF password IS NOT EMPTY AND NOT MATCHES_PATTERN(password,
PASSWORD_REGEX) THEN
        errors.ADD("Password must be at least 8 characters and contain letters and numbers")
    END IF

    // Step 5: Handle validation outcome
    IF errors IS EMPTY THEN
        SANITIZE(name, email, password)
        SAVE_TO_DATABASE(name, email, password)
        DISPLAY "Form submitted successfully"
    ELSE
        FOR EACH error IN errors DO
            DISPLAY error
        END FOR
        DISPLAY "Please correct the errors and resubmit"
    END IF

END IF
END
```